ELITE IGCSE ACADEMY

Student Past Paper Solutions

Jan 2021 P1H Worked Solutions

Jan 2021 | P1H

29 questions ▪ question image followed by worked solution

PREPARED BY

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STUDENT EDITION
Worked answers included

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PAST PAPER QUESTION

Q#: 1 Marks: 4

TOPIC: **Ratio Problem Solving**

Jan 2021 P1H - Jan 2021 | P1H

1 Pieter owns a currency conversion shop.

Last Monday, Pieter changed a total of 20 160 rand into a number of different currencies.

He changed $\frac{3}{10}$ of the 20 160 rand into euros.

He changed the rest of the rands into dollars, rupees and francs in the ratios $9:5:2$

Pieter changed more rands into dollars than he changed into francs.

Work out how many more.

.... rand

(Total for Question 1 is 4 marks)

PAST PAPER SOLUTION

Q#: 1 Marks: 4

TOPIC: **Ratio Problem Solving**

Jan 2021 P1H - Jan 2021 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Corrected to Ratio Problem Solving. The currency words are incidental; the main skill is ratio sharing.

Method

$\frac{3}{10}$ of the money is changed to euros, so the rest is

$$\frac{7}{10} \times 20160 = 14112$$

This rest is divided in the ratio

$$\text{dollars} : \text{rupees} : \text{francs} = 9 : 5 : 2$$

There are

$$9 + 5 + 2 = 16 \text{ parts}$$

The difference between dollars and francs is

$$9 - 2 = 7 \text{ parts}$$

So the difference is

$$\frac{7}{16} \times 14112 = 6174$$

FINAL ANSWER

6174 rand more is changed into dollars than francs.

PAST PAPER QUESTION

Q#: 2 Marks: 5

TOPIC: **Statistics Toolkit**

Jan 2021 P1H - Jan 2021 | P1H

2 The table gives information about the speeds, in kilometres per hour, of 80 motorbikes as each pass under a bridge.

Speed (s kilometres per hour)	**Frequency**
$40 < s \leqslant 50$	10
$50 < s \leqslant 60$	16
$60 < s \leqslant 70$	19
$70 < s \leqslant 80$	23
$80 < s \leqslant 90$	12

(a) Write down the modal class.

....

(1)

(b) Work out an estimate for the mean speed of the motorbikes as they pass under the bridge. Give your answer correct to 3 significant figures.

.... kilometres per hour

(4)

(Total for Question 2 is 5 marks)

PAST PAPER SOLUTION

Q#: 2 Marks: 5

TOPIC: **Statistics Toolkit**

Jan 2021 P1H - Jan 2021 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Statistics Toolkit. The tag is correct.

Solution

(a) The greatest frequency is 23, so the modal class is

$$70 < s \leq 80$$

(b) Use midpoints $45, 55, 65, 75, 85$:

$$\frac{45(10) + 55(16) + 65(19) + 75(23) + 85(12)}{80} = 66.375$$

To 3 significant figures:

$$66.4$$

FINAL ANSWER

(a) $70 < s \leq 80$, (b) 66.4 km/h.

PAST PAPER QUESTION

Q#: 3 Marks: 4

TOPIC: **Volume & Surface Area**

Jan 2021 P1H - Jan 2021 | P1H

3 The diagram shows a container for water in the shape of a prism.

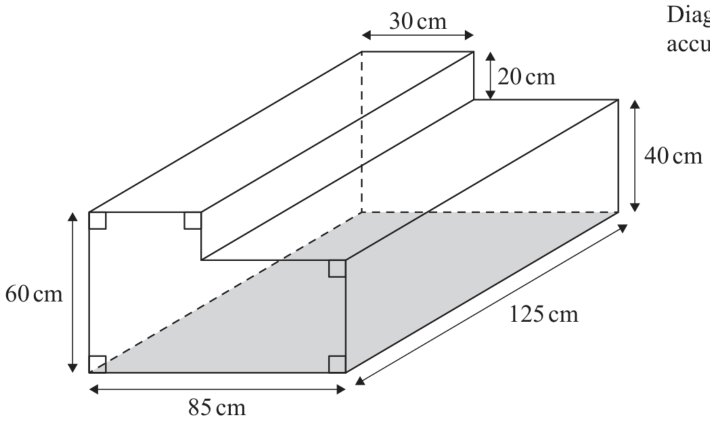

Diagram **NOT** accurately drawn

The rectangular base of the prism, shown shaded in the diagram, is horizontal.
The container is completely full of water.

Tuah is going to use a pump to empty the water from the container so that the volume of water in the container decreases at a constant rate.

The pump starts to empty water from the container at 10 30 and at 12 00 the water level in the container has dropped by 20 cm.

Find the time at which all the water has been pumped out of the container.

....

(Total for Question 3 is 4 marks)

PAST PAPER SOLUTION Q#: 3 Marks: 4

TOPIC: **Volume & Surface Area**

Jan 2021 P1H - Jan 2021 | P1H

WORKED SOLUTION

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Topic check. Corrected to Volume & Surface Area. The problem is about the volume of a prism being emptied at a constant rate.

Method

The cross-section is an L-shape.
Area of the lower rectangle:

$$85 \times 40 = 3400$$

Area of the upper rectangle:

$$30 \times 20 = 600$$

So the cross-sectional area is

$$3400 + 600 = 4000 \text{ cm}^2$$

The full volume is

$$4000 \times 125 = 500000 \text{ cm}^3$$

In the first 1.5 hours, the water level drops by 20 cm. This is the top rectangular part:

$$30 \times 20 \times 125 = 75000 \text{ cm}^3$$

So the pumping rate is

$$\frac{75000}{1.5} = 50000 \text{ cm}^3/\text{hour}$$

Time to empty the whole container:

$$\frac{500000}{50000} = 10 \text{ hours}$$

Starting at 10:30, ten hours later is

$$20{:}30$$

FINAL ANSWER

20:30

PAST PAPER QUESTION Q#: 4 Marks: 2

TOPIC: **Set Notation & Venn Diagrams**

Jan 2021 P1H - Jan 2021 | P1H

4 $\mathscr{E} = \{20, 21, 22, 23, 24, 25, 26, 27, 28, 29\}$

$A = \{\text{odd numbers}\}$
$B = \{\text{multiples of } 3\}$

List the members of the set

(i) $A \cap B$

....

(1)

(ii) $A \cup B$

....

(1)

(Total for Question 4 is 2 marks)

PAST PAPER SOLUTION

Q#: 4 Marks: 2

TOPIC: **Set Notation & Venn Diagrams**

Jan 2021 P1H - Jan 2021 | P1H

WORKED SOLUTION

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Topic check. Set notation and Venn diagrams. The tag is correct.

Method

$$\mathcal{E} = \{20, 21, 22, 23, 24, 25, 26, 27, 28, 29\}$$

The odd numbers are

$$A = \{21, 23, 25, 27, 29\}$$

The multiples of 3 are

$$B = \{21, 24, 27\}$$

The numbers in both sets are

$$A \cap B = \{21, 27\}$$

The numbers in either set are

$$A \cup B = \{21, 23, 24, 25, 27, 29\}$$

FINAL ANSWER

(i) $\{21, 27\}$, (ii) $\{21, 23, 24, 25, 27, 29\}$

PAST PAPER QUESTION Q#: 5 Marks: 5

TOPIC: **Solving Linear Equations**

Jan 2021 P1H - Jan 2021 | P1H

5 (a) Factorise fully $15y^4 + 20uy^3$

....

(2)

(b) Solve $4 - 3x = \dfrac{5 - 8x}{4}$

Show clear algebraic working.

$x =$ ....

(3)

(Total for Question 5 is 5 marks)

PAST PAPER SOLUTION

Q#: 5 Marks: 5

TOPIC: **Solving Linear Equations**

Jan 2021 P1H - Jan 2021 | P1H

WORKED SOLUTION

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Topic check. Corrected to solving linear equations. The original tag was factorising, but the equation is the larger part.

Solution

(a)

$$15y^4 + 20uy^3 = 5y^3(3y + 4u)$$

(b)

$$4 - 3x = \frac{5 - 8x}{4}$$

$$16 - 12x = 5 - 8x$$

$$11 = 4x$$

FINAL ANSWER

(a) $5y^3(3y + 4u)$, (b) $x = \frac{11}{4}$

PAST PAPER QUESTION

Q#: 6 **Marks: 2**

TOPIC: **Powers, Roots & Standard Form**

Jan 2021 P1H - Jan 2021 | P1H

6 (a) Write 2 840 000 000 in standard form.

....

(1)

(b) Write 2.5×10^{-4} as an ordinary number.

....

(1)

(Total for Question 6 is 2 marks)

PAST PAPER SOLUTION

Q#: 6 Marks: 2

TOPIC: **Powers, Roots & Standard Form**

Jan 2021 P1H - Jan 2021 | P1H

WORKED SOLUTION

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Topic check. Powers, roots and standard form. The tag is correct.

Method

$$2840000000 = 2.84 \times 10^9$$

Also,

$$2.5 \times 10^{-4} = 0.00025$$

FINAL ANSWER

(a) 2.84×10^9, (b) 0.00025

PAST PAPER QUESTION

Q#: 7 Marks: 6

TOPIC: **Compound Interest & Depreciation**

Jan 2021 P1H - Jan 2021 | P1H

7 Chen invests 40 000 yuan in a fixed-term bond for 3 years.

The fixed-term bond pays compound interest at a rate of 3.5% each year.

(a) Work out the value of Chen's investment at the end of 3 years.
Give your answer to the nearest yuan.

.... yuan
(3)

Wang invested P yuan.
The value of his investment decreased by 6.5% each year.

At the end of the first year, the value of Wang's investment was 30 481 yuan.

(b) Work out the value of P.

$P =$ ....
(3)

(Total for Question 7 is 6 marks)

PAST PAPER SOLUTION

Q#: 7 Marks: 6

TOPIC: **Compound Interest & Depreciation**

Jan 2021 P1H - Jan 2021 | P1H

WORKED SOLUTION

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Topic check. Compound Interest and Depreciation. The tag is correct.

Method

For part (a),

$$40000(1.035)^3 = 44348.715$$

Correct to the nearest yuan,

$$44348.715 \approx 44349$$

For part (b), after one year the value is 93.5% of P.

$$0.935P = 30481$$

$$P = \frac{30481}{0.935} = 32600$$

FINAL ANSWER

(a) 44349 yuan, (b) $P = 32600$

PAST PAPER QUESTION

TOPIC: **Circles, Arcs & Sectors**

Jan 2021 P1H - Jan 2021 | P1H

Q#: 8 Marks: 3

8 The region, shown shaded in the diagram, is a path.

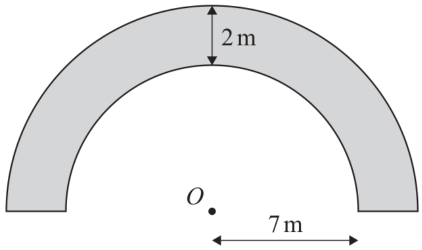

Diagram **NOT** accurately drawn

The boundary of the path is formed by two semicircles, with the same centre O, and two straight lines.

The inner semicircle has a radius of 7 metres.
The path has a width of 2 metres.

Work out the perimeter of the path.
Give your answer correct to one decimal place.

.... m

(Total for Question 8 is 3 marks)

PAST PAPER SOLUTION

Q#: 8 Marks: 3

TOPIC: **Circles, Arcs & Sectors**

Jan 2021 P1H - Jan 2021 | P1H

WORKED SOLUTION

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Topic check. Corrected to circles, arcs and sectors. The question is the perimeter of a path made from semicircles.

Solution

The inner radius is 7 m.
The path is 2 m wide, so the outer radius is

$$7 + 2 = 9 \text{ m}$$

Perimeter of the path:

$$\text{inner semicircle} + \text{outer semicircle} + 2 \text{ straight ends}$$

$$= 7\pi + 9\pi + 2 + 2$$

$$= 16\pi + 4$$

$$= 54.265\ldots$$

FINAL ANSWER

54.3 m.

PAST PAPER QUESTION

Q#: 9 Marks: 5

Topic: **Factorising**

Jan 2021 P1H - Jan 2021 | P1H

9 (a) Simplify $(2x^3y^5)^4$

....

(2)

(b) (i) Factorise $x^2 + 5x - 36$

....

(2)

(ii) Hence, solve $x^2 + 5x - 36 = 0$

....

(1)

(Total for Question 9 is 5 marks)

PAST PAPER SOLUTION

Q#: 9 Marks: 5

TOPIC: **Factorising**

Jan 2021 P1H - Jan 2021 | P1H

WORKED SOLUTION

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Topic check. Factorising. The tag is correct.

Solution

(a)

$$(2x^3y^5)^4 = 16x^{12}y^{20}$$

(b)(i)

$$x^2 + 5x - 36 = (x + 9)(x - 4)$$

(b)(ii) $x = -9$ or $x = 4$.

FINAL ANSWER

(a) $16x^{12}y^{20}$, (b)(i) $(x + 9)(x - 4)$, (b)(ii) $x = -9$ or $x = 4$

PAST PAPER QUESTION

Q#: 10 Marks: 4

TOPIC: **Right-Angled Triangles - Pythagoras & Trigonometry**

Jan 2021 P1H - Jan 2021 | P1H

10 Here is isosceles triangle *ABC*.

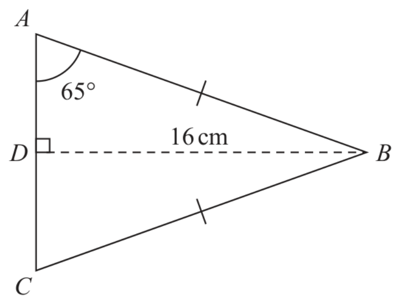

Diagram **NOT** accurately drawn

D is the midpoint of *AC* and *DB* = 16 cm.

Angle *DAB* = 65°

Work out the perimeter of triangle *ABC*.
Give your answer correct to one decimal place.

.... cm

(Total for Question 10 is 4 marks)

PAST PAPER SOLUTION

Q#: 10 Marks: 4

TOPIC: **Right-Angled Triangles - Pythagoras & Trigonometry**

Jan 2021 P1H - Jan 2021 | P1H

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Topic check. Corrected to right-angled trigonometry. The solution uses right-triangle trigonometry in an isosceles triangle.

Solution

Since ABC is isosceles and D is the midpoint of AC, BD is perpendicular to AC.
In right triangle ABD,

$$BD = 16, \qquad \angle DAB = 65^\circ$$

$$\sin 65^\circ = \frac{16}{AB}$$

$$AB = \frac{16}{\sin 65^\circ}$$

Also,

$$\tan 65^\circ = \frac{16}{AD}$$

$$AD = \frac{16}{\tan 65^\circ}$$

The perimeter is

$$AB + BC + AC = 2AB + 2AD$$

$$= 2\left(\frac{16}{\sin 65^\circ}\right) + 2\left(\frac{16}{\tan 65^\circ}\right)$$

$$= 50.2299\ldots$$

FINAL ANSWER

50.2 cm.

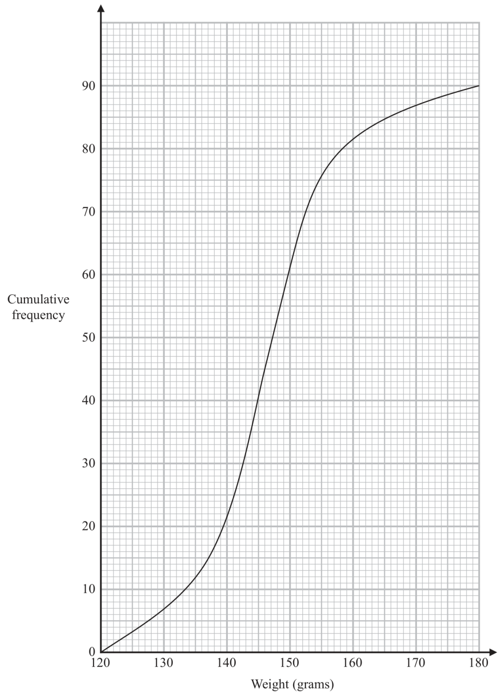
PAST PAPER QUESTION

Q#: 11 **Marks: 5**

TOPIC: **Cumulative Frequency Diagrams**

Jan 2021 P1H - Jan 2021 | P1H

11 The cumulative frequency graph gives information about the weights, in grams, of 90 bags of sweets.



(a) Find an estimate for the median of the weights of these bags of sweets.

.... grams

(2)

Roberto sells the bags of sweets to raise money for charity.
Bags with a weight greater than *d* grams are labelled large bags and sold for 3.75 euros each bag.

The total amount of money he receives by selling all the large bags is 93.75 euros.

(b) Find the value of *d*.

d = ...

(3)

(Total for Question 11 is 5 marks)

PAST PAPER SOLUTION

Q#: 11 Marks: 5

TOPIC: **Cumulative Frequency Diagrams**

Jan 2021 P1H - Jan 2021 | P1H

WORKED SOLUTION

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Topic check. Cumulative Frequency Diagrams. The tag is correct.

Solution

There are 90 bags, so the median is at cumulative frequency 45.
From the graph,

$$\text{median} \approx 146 \text{ g}$$

Large bags sold:

$$\frac{93.75}{3.75} = 25$$

So 25 bags have weight greater than d.
That means

$$90 - 25 = 65$$

bags have weight d grams or less.
From the graph, cumulative frequency 65 gives

$$d \approx 150$$

FINAL ANSWER
median about 146 g, and $d \approx 150$.

PAST PAPER QUESTION

Q#: 12 Marks: 6

TOPIC: **Expanding Brackets**

Jan 2021 P1H - Jan 2021 | P1H

12 (a) Express $\dfrac{4}{x-2} - \dfrac{3}{x+1}$ as a single fraction.

Give your answer in its simplest form.

....

(3)

(b) Expand and simplify $2x(x-5)(x-3)$

....

(3)

(Total for Question 12 is 6 marks)

PAST PAPER SOLUTION

Q#: 12 Marks: 6

TOPIC: **Expanding Brackets**

Jan 2021 P1H - Jan 2021 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Expanding brackets. The tag is acceptable.

Solution

(a)

$$\frac{4}{x-2} - \frac{3}{x+1} = \frac{4(x+1) - 3(x-2)}{(x-2)(x+1)}$$
$$= \frac{4x+4-3x+6}{(x-2)(x+1)} = \frac{x+10}{(x-2)(x+1)}$$

(b)

$$2x(x-5)(x-3) = 2x(x^2 - 8x + 15)$$

$$= 2x^3 - 16x^2 + 30x$$

FINAL ANSWER

(a) $\frac{x+10}{(x-2)(x+1)}$, (b) $2x^3 - 16x^2 + 30x$

PAST PAPER QUESTION

Q#: 13 **Marks: 3**

TOPIC: **Coordinate Geometry**

Jan 2021 P1H - Jan 2021 | P1H

13 Point A has coordinates (5, 8)
Point B has coordinates (9, –4)

(a) Work out the gradient of AB.

....

(2)

The straight line **L** has equation $y = -4x + 5$

(b) Write down the gradient of a straight line that is perpendicular to **L**.

....

(1)

(Total for Question 13 is 3 marks)

PAST PAPER SOLUTION

Q#: 13 Marks: 3

TOPIC: **Coordinate Geometry**

Jan 2021 P1H - Jan 2021 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Coordinate geometry. The tag is correct.

Solution

(a) The gradient of AB, where $A(5, 8)$ and $B(9, -4)$, is

$$\frac{-4-8}{9-5} = \frac{-12}{4} = -3$$

(b) A line perpendicular to a line with gradient -3 has gradient

$$\frac{1}{3}$$

FINAL ANSWER

Gradient of $AB = -3$, perpendicular gradient $\frac{1}{3}$.

PAST PAPER QUESTION

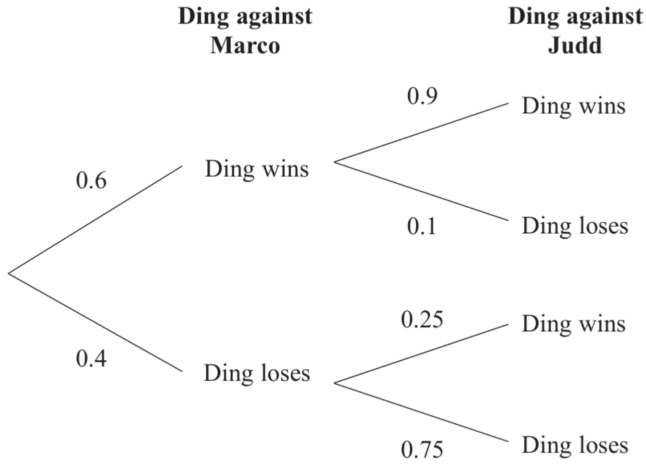
Q#: 14 Marks: 5

TOPIC: **Probability Diagrams - Venn & Tree Diagrams**

Jan 2021 P1H - Jan 2021 | P1H

14 Ding is going to play one game of snooker against each of two of his friends, Marco and Judd.

The probability tree diagram gives information about the probabilities that Ding will win or lose each of these two games.



(a) Work out the probability that Ding will win both games.

....

(2)

(b) Work out the probability that Ding will win exactly one of the games.

....

(3)

(Total for Question 14 is 5 marks)

PAST PAPER SOLUTION

Q#: 14 Marks: 5

TOPIC: **Probability Diagrams - Venn & Tree Diagrams**

Jan 2021 P1H - Jan 2021 | P1H

WORKED SOLUTION

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Topic check. Probability Diagrams - Venn & Tree Diagrams. The tag is correct.

Solution

Winning both frames:

$$0.6 \times 0.9 = 0.54$$

Winning exactly one frame:

$$0.6 \times 0.1 + 0.4 \times 0.25$$

$$= 0.06 + 0.10 = 0.16$$

FINAL ANSWER

0.54, 0.16.

PAST PAPER QUESTION

Q#: 15 **Marks: 3**

TOPIC: **Rounding, Estimation & Bounds**

Jan 2021 P1H - Jan 2021 | P1H

15

$$a = \frac{v - u}{t}$$

v = 9.6 correct to 1 decimal place
u = 3.8 correct to 1 decimal place
t = 1.84 correct to 2 decimal places

Calculate the upper bound for the value of a.
Give your answer as a decimal correct to 2 decimal places.
Show your working clearly.

....

(Total for Question 15 is 3 marks)

PAST PAPER SOLUTION

Q#: 15 Marks: 3

TOPIC: **Rounding, Estimation & Bounds**

Jan 2021 P1H - Jan 2021 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Rounding, Estimation and Bounds. The tag is correct.

Method

$$a = \frac{v-u}{t}$$

For the upper bound, use the upper bound of v, the lower bound of u, and the lower bound of t.

$$v < 9.65, \quad u \geq 3.75, \quad t \geq 1.835$$

$$a_{\text{upper}} = \frac{9.65 - 3.75}{1.835} = 3.2152\ldots$$

FINAL ANSWER

3.22

PAST PAPER QUESTION

Q#: 16 Marks: 5

TOPIC: **Bearings, Scale Drawing & Constructions**

Jan 2021 P1H - Jan 2021 | P1H

16 The diagram shows the positions of three ships, A, B and C.

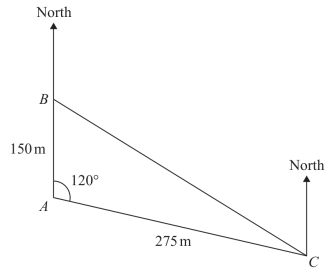

Diagram **NOT** accurately drawn

Ship B is due north of ship A.

The bearing of ship C from ship A is 120°

Calculate the bearing of ship C from ship B.
Give your answer correct to the nearest degree.

.... °

(Total for Question 16 is 5 marks)

PAST PAPER SOLUTION

Q#: 16 Marks: 5

TOPIC: **Bearings, Scale Drawing & Constructions**

Jan 2021 P1H - Jan 2021 | P1H

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Topic check. Bearings, Scale Drawing & Constructions. The tag is correct.

Solution

From A, ship C is on a bearing of $120°$, and $AC = 275$ m.
Resolve AC:

$$\text{east component} = 275 \sin 120° = 238.16\ldots$$

$$\text{north component} = 275 \cos 120° = -137.5$$

Ship B is 150 m north of A, so from B to C:

$$\text{east} = 238.16\ldots, \qquad \text{north} = -137.5 - 150 = -287.5$$

The bearing is therefore in the south-east quadrant:

$$180° - \tan^{-1}\left(\frac{238.16\ldots}{287.5}\right) = 140.36\ldots°$$

FINAL ANSWER

$140°$.

PAST PAPER QUESTION Q#: 17 Marks: 5

TOPIC: **Differentiation**

Jan 2021 P1H - Jan 2021 | P1H

17 A solid, **S**, is made from a hemisphere and a cylinder.

The centre of the circular face of the hemisphere and the centre of the top face of the cylinder are at the same point.

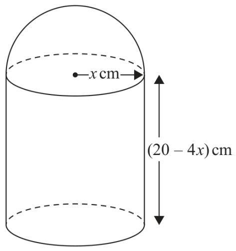

Diagram **NOT** accurately drawn

The radius of the cylinder and the radius of the hemisphere are both x cm.
The height of the cylinder is $(20 - 4x)$ cm.

The volume of **S** is V cm^3 where $V = \frac{1}{3}\pi y$

Find the maximum value of y.
Show clear algebraic working.

....

(Total for Question 17 is 5 marks)

PAST PAPER SOLUTION

Q#: 17 Marks: 5

TOPIC: **Differentiation**

Jan 2021 P1H - Jan 2021 | P1H

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Topic check. Correct. This is maximisation using differentiation.

Solution

Volume of the solid is cylinder plus hemisphere:

$$V = \pi x^2(20 - 4x) + \frac{2}{3}\pi x^3$$

$$V = 20\pi x^2 - 4\pi x^3 + \frac{2}{3}\pi x^3$$

$$V = 20\pi x^2 - \frac{10}{3}\pi x^3$$

The question gives

$$V = \frac{1}{3}\pi y$$

So

$$y = 60x^2 - 10x^3$$

Differentiate:

$$\frac{dy}{dx} = 120x - 30x^2$$

For a maximum,

$$120x - 30x^2 = 0$$

$$30x(4 - x) = 0$$

So the valid value is

$$x = 4$$

$$y = 60(4^2) - 10(4^3) = 960 - 640 = 320$$

FINAL ANSWER

The maximum value of y is 320.

PAST PAPER QUESTION Q#: 18 Marks: 3

TOPIC: **Surds**

Jan 2021 P1H - Jan 2021 | P1H

18 Given that $(8 - \sqrt{x})(5 + \sqrt{x}) = y\sqrt{x} + 21$ where x is a prime number and y is an integer,

find the value of x and the value of y.
Show each stage of your working clearly.

$x =$ ....

$y =$ ....

(Total for Question 18 is 3 marks)

PAST PAPER SOLUTION

Q#: 18 **Marks: 3**

TOPIC: **Surds**

Jan 2021 P1H - Jan 2021 | P1H

WORKED SOLUTION

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Topic check. Corrected to surds. The question uses surd expansion and coefficient matching.

Solution

$$(8-\sqrt{x})(5+\sqrt{x}) = 40 + 8\sqrt{x} - 5\sqrt{x} - x$$

$$= 40 + 3\sqrt{x} - x$$

This is equal to

$$y\sqrt{x} + 21$$

Compare the rational parts:

$$40 - x = 21$$

$$x = 19$$

Compare the surd coefficients:

$$y = 3$$

FINAL ANSWER

$x = 19$, $y = 3$

PAST PAPER QUESTION

Q#: 19 Marks: 5

TOPIC: **Simultaneous Equations**

Jan 2021 P1H - Jan 2021 | P1H

19 Solve the simultaneous equations

$$x^2 - 9y - x = 2y^2 - 12$$
$$x + 2y - 1 = 0$$

Show clear algebraic working.

....

(Total for Question 19 is 5 marks)

PAST PAPER SOLUTION

Q#: 19 Marks: 5

TOPIC: **Simultaneous Equations**

Jan 2021 P1H - Jan 2021 | P1H

WORKED SOLUTION

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Topic check. Simultaneous equations. The tag is correct.

Solution

$$x^2 - 9y - x = 2y^2 - 12$$

$$x + 2y - 1 = 0$$

From the second equation:

$$x = 1 - 2y$$

Substitute into the first equation:

$$(1 - 2y)^2 - 9y - (1 - 2y) = 2y^2 - 12$$

$$4y^2 - 11y = 2y^2 - 12$$

$$2y^2 - 11y + 12 = 0$$

Factorise:

$$(2y - 3)(y - 4) = 0$$

So

$$y = \frac{3}{2} \quad \text{or} \quad y = 4$$

If $y = \frac{3}{2}$:

$$x = 1 - 2\left(\frac{3}{2}\right) = -2$$

If $y = 4$:

$$x = 1 - 8 = -7$$

FINAL ANSWER

$(x, y) = \left(-2, \frac{3}{2}\right)$ or $(-7, 4)$.

PAST PAPER QUESTION

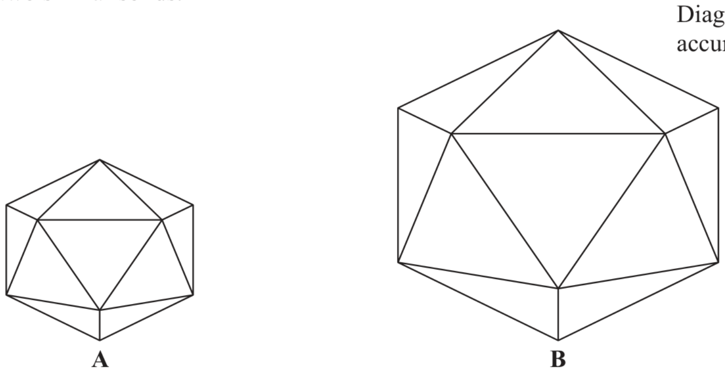
Q#: 20 Marks: 3

TOPIC: **Area & Volume of Similar Shapes**

Jan 2021 P1H - Jan 2021 | P1H

20 **A** and **B** are two similar solids.

Diagram **NOT** accurately drawn



A has a volume of 1836 cm^3
B has a volume of 4352 cm^3

B has a total surface area of 1120 cm^2

Work out the total surface area of **A**.

.... cm^2

(Total for Question 20 is 3 marks)

PAST PAPER SOLUTION

Q#: 20 Marks: 3

TOPIC: **Area & Volume of Similar Shapes**

Jan 2021 P1H - Jan 2021 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

The volume ratio of solid A to solid B is

$$1836 : 4352$$

Simplify:

$$\frac{1836}{4352} = \frac{27}{64}$$

So the linear ratio is

$$\sqrt[3]{27} : \sqrt[3]{64} = 3 : 4$$

Surface area ratio:

$$3^2 : 4^2 = 9 : 16$$

Solid B has surface area 1120 cm^2, so solid A has surface area

$$1120 \times \frac{9}{16} = 630$$

FINAL ANSWER

$\boxed{630 \text{ cm}^2}$

PAST PAPER QUESTION Q#: 20 Marks: 3

Topic: **Area & Volume of Similar Shapes**

Jan 2021 P1H - Jan 2021 | P1H

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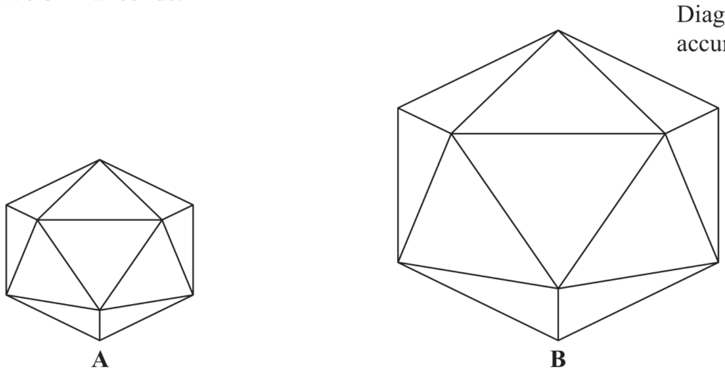

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PAST PAPER SOLUTION Q#: 20 Marks: 3

TOPIC: **Area & Volume of Similar Shapes**

Jan 2021 P1H - Jan 2021 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

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$$1120 \times \frac{9}{16} = 630$$

FINAL ANSWER

$\boxed{630 \text{ cm}^2}$

PAST PAPER QUESTION Q#: 21 Marks: 4

TOPIC: **Transformations of Graphs**

Jan 2021 P1H - Jan 2021 | P1H

21 A curve has equation $y = f(x)$

The coordinates of the minimum point on this curve are (–9, 15)

(a) Write down the coordinates of the minimum point on the curve with equation

(i) $y = f(x + 3)$

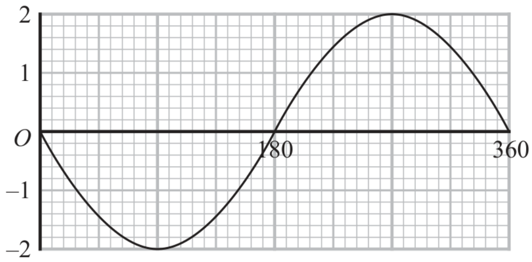
(.......................... , ...)

(ii) $y = \frac{1}{3}f(x)$

(.......................... , ...)

(2)

The graph of $y = a\cos(x + b)°$ for $0 \leqslant x \leqslant 360$ is drawn on the grid below.



Given that $a > 0$ and that $0 < b < 360$

(b) find the value of a and the value of b.

a = ...

b = ...

(2)

(Total for Question 21 is 4 marks)

PAST PAPER SOLUTION

Q#: 21 Marks: 4

TOPIC: **Transformations of Graphs**

Jan 2021 P1H - Jan 2021 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

The minimum point of $y = f(x)$ is $(-9, 15)$.
For

$$y = f(x + 3)$$

the graph moves 3 units left:

$$(-9, 15) \to (-12, 15)$$

For

$$y = \frac{1}{3} f(x)$$

the y-coordinate is divided by 3:

$$(-9, 15) \to (-9, 5)$$

For part (b), the cosine graph has maximum 2 and minimum -2, so

$$a = 2$$

The graph has a minimum at $x = 90°$. For $a \cos(x + b)°$, a minimum occurs when

$$x + b = 180$$

So

$$90 + b = 180$$

$$b = 90$$

FINAL ANSWER

$(-12, 15)$, $(-9, 5)$, $a = 2$, $b = 90$

PAST PAPER QUESTION

Q#: 21 Marks: 4

TOPIC: **Transformations of Graphs**

Jan 2021 P1H - Jan 2021 | P1H

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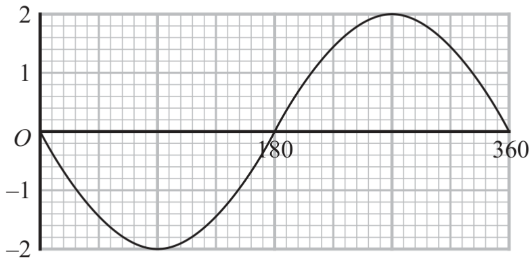

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(Total for Question 21 is 4 marks)

PAST PAPER SOLUTION Q#: 21 Marks: 4

TOPIC: **Transformations of Graphs**

Jan 2021 P1H - Jan 2021 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

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FINAL ANSWER
$(-12, 15)$, $(-9, 5)$, $a = 2$, $b = 90$

PAST PAPER QUESTION

TOPIC: **Functions**

Jan 2021 P1H - Jan 2021 | P1H

Q#: 22 Marks: 3

22 The function f is such that $f(x) = x^2 - 8x + 5$ where $x \leqslant 4$

Express the inverse function f^{-1} in the form $f^{-1}(x) = \ldots$

$f^{-1}(x) =$ ....

(Total for Question 22 is 3 marks)

PAST PAPER SOLUTION

Q#: 22 Marks: 3

TOPIC: **Functions**

Jan 2021 P1H - Jan 2021 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

$$f(x) = x^2 - 8x + 5$$

$$f(x) = (x-4)^2 - 11$$

Let

$$y = (x-4)^2 - 11$$

$$y + 11 = (x-4)^2$$

Since the domain is $x < 4$, take the negative square root:

$$x - 4 = -\sqrt{y+11}$$

$$x = 4 - \sqrt{y+11}$$

FINAL ANSWER

$f^{-1}(x) = 4 - \sqrt{x+11}$

PAST PAPER QUESTION Q#: 22 Marks: 3

TOPIC: **Functions**

Jan 2021 P1H - Jan 2021 | P1H

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$f^{-1}(x) = \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots$

(Total for Question 22 is 3 marks)

PAST PAPER SOLUTION

Q#: 22 Marks: 3

TOPIC: **Functions**

Jan 2021 P1H - Jan 2021 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

$$f(x) = x^2 - 8x + 5$$

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FINAL ANSWER

$f^{-1}(x) = 4 - \sqrt{x+11}$

PAST PAPER QUESTION

Q#: 23 Marks: 6

TOPIC: **Vectors**

Jan 2021 P1H - Jan 2021 | P1H

23 *OAB* is a triangle.

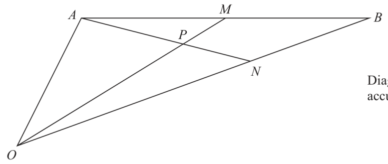


Diagram **NOT** accurately drawn

$\overrightarrow{OA} = 2\mathbf{a}$ and $\overrightarrow{OB} = 2\mathbf{b}$

M is the midpoint of *AB*.
N is the point on *OB* such that $ON : NB = 2 : 1$

P is the point on *AN* such that *OPM* is a straight line.

Use a vector method to find $OP : PM$
Show your working clearly.

....

(Total for Question 23 is 6 marks)

PAST PAPER SOLUTION Q#: 23 Marks: 6

TOPIC: **Vectors**

Jan 2021 P1H - Jan 2021 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Moved from Coordinate Geometry to Vectors.

Method

$$\vec{OA} = 2\mathbf{a}, \qquad \vec{OB} = 2\mathbf{b}$$

Since M is the midpoint of AB,

$$\vec{OM} = \frac{2\mathbf{a} + 2\mathbf{b}}{2}$$

$$\vec{OM} = \mathbf{a} + \mathbf{b}$$

Since $ON : NB = 2 : 1$,

$$\vec{ON} = \frac{2}{3}\vec{OB}$$

$$\vec{ON} = \frac{2}{3}(2\mathbf{b}) = \frac{4}{3}\mathbf{b}$$

Point P lies on OM, so

$$\vec{OP} = t(\mathbf{a} + \mathbf{b})$$

Point P also lies on AN, so

$$\vec{OP} = 2\mathbf{a} + s\left(\frac{4}{3}\mathbf{b} - 2\mathbf{a}\right)$$

$$\vec{OP} = (2 - 2s)\mathbf{a} + \frac{4s}{3}\mathbf{b}$$

Equate coefficients of $\mathbf{a}$ and $\mathbf{b}$:

$$t = 2 - 2s$$

$$t = \frac{4s}{3}$$

So

$$\frac{4s}{3} = 2 - 2s$$

$$4s = 6 - 6s$$

$$10s = 6$$

$$s = \frac{3}{5}$$

Then

$$t = \frac{4}{5}$$

So P is $\frac{4}{5}$ of the way from O to M.

FINAL ANSWER
$OP : PM = 4 : 1$

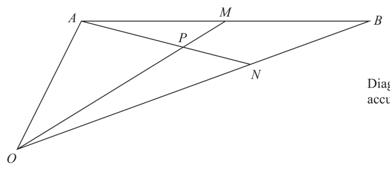
PAST PAPER QUESTION

Q#: 23 Marks: 6

TOPIC: **Vectors**

Jan 2021 P1H - Jan 2021 | P1H

23 OAB is a triangle.



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Use a vector method to find $OP : PM$
Show your working clearly.

....

(Total for Question 23 is 6 marks)

PAST PAPER SOLUTION

Q#: 23 Marks: 6

Topic: **Vectors**

Jan 2021 P1H - Jan 2021 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Moved from Coordinate Geometry to Vectors.

Method

$$\vec{OA} = 2\mathbf{a}, \qquad \vec{OB} = 2\mathbf{b}$$

Since M is the midpoint of AB,

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$$\vec{OM} = \mathbf{a} + \mathbf{b}$$

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So

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So P is $\frac{4}{5}$ of the way from O to M.

FINAL ANSWER
$OP : PM = 4 : 1$

PAST PAPER QUESTION

Q#: 24 Marks: 4

TOPIC: **Sequences**

Jan 2021 P1H - Jan 2021 | P1H

24 An arithmetic series has first term a and common difference d.

The sum of the first $2n$ terms of the series is four times the sum of the first n terms of the series.

Find an expression for a in terms of d.
Show your working clearly.

$a =$ ....

(Total for Question 24 is 4 marks)

TOTAL FOR PAPER IS 100 MARKS

PAST PAPER SOLUTION

Q#: 24 Marks: 4

TOPIC: **Sequences**

Jan 2021 P1H - Jan 2021 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

$$S_n = \frac{n}{2}(2a + (n-1)d)$$

$$S_{2n} = n(2a + (2n-1)d)$$

Given

$$S_{2n} = 4S_n$$

$$n(2a + (2n-1)d) = 4 \cdot \frac{n}{2}(2a + (n-1)d)$$

Divide by n:

$$2a + (2n-1)d = 2(2a + (n-1)d)$$

$$2a + 2nd - d = 4a + 2nd - 2d$$

$$d = 2a$$

$$a = \frac{d}{2}$$

FINAL ANSWER

$a = \frac{d}{2}$

PAST PAPER QUESTION

Q#: 24 Marks: 4

TOPIC: **Sequences**

Jan 2021 P1H - Jan 2021 | P1H

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Jan 2021 P1H - Jan 2021 | P1H

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FINAL ANSWER

$a = \frac{d}{2}$